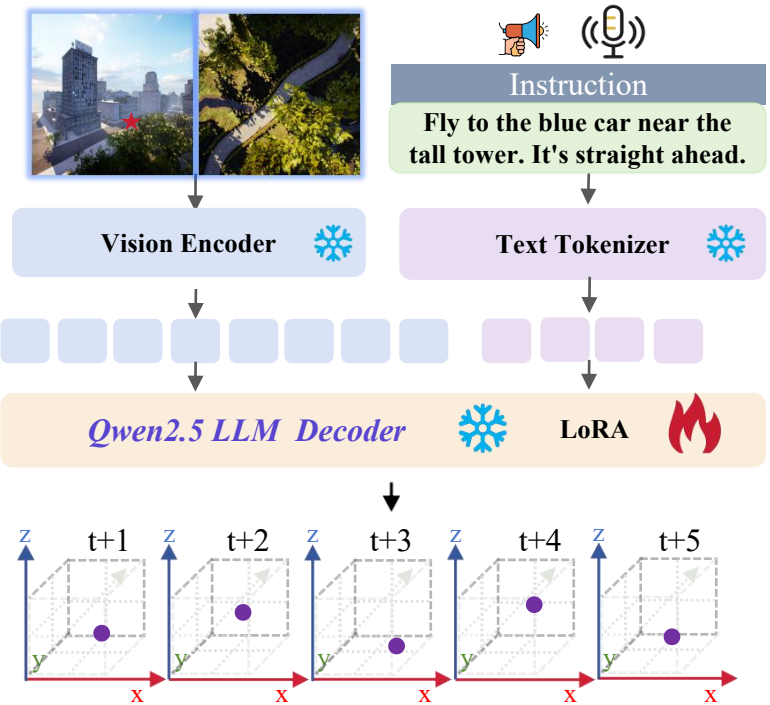


TRAINING STAGE



3DG-VLN learns to predict future 3D waypoints from a high-level instruction, high-resolution dual-view observations and a coarse 3D direction.

INFERENCE STAGE

Observations



1) Instruction Processing

Instruction: Fly to the black car. It's off to the front-left.



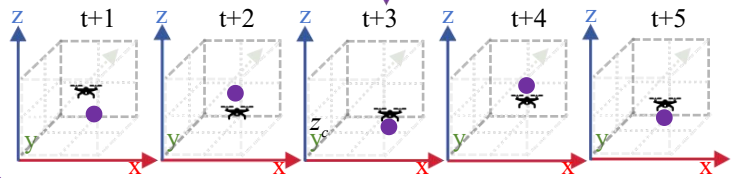
Target Object & Attribute

2) Waypoint Prediction



Qwen2.5-VL (Waypoint Predictor)

Predicted Future 3D Waypoints (5-step sequence)



UAV Execution



New Observations

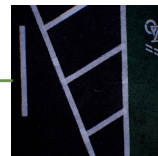


3) Online 3D Direction Updating

3.1 Open-Vocabulary Detection



Open-Set Detector



3.2 Validity Determination

Valid?

No Do Not Update Direction

Yes

3.3 Direction Determination

front-left front-right down front

Down-view Valid?

Yes

No

Front-view Geometric Processing

